foods such as fruits and vegetables normally contain same micro organisms and may be contaminated with additional organisms during handling. Many factors that influence the growth of the micro organisms in food some of the factors are intrinsic and some others are extrinsic factors. Spoilage of food can be defined as any visible or invisible change which can make food or product from food unacceptable for human consumption spoilage of food not only causes health hazard to the consumer but also cause large economic losses. Food poisoning refers to the toxicity introduced into food by micro organism and their product. Food intoxication or food poisoning results from ingestion of foods containing performed microbial toxins. Foods can be preserved by a variety of methods. It is vital to eliminate or reduce the populations of spoilage and disease causing micro organisms and to maintain the micro biological quality of a food with proper storage and packaging.

Food borne disease has been defined by the world health organization (WHO) as a disease of an infectious or toxic nature caused by or thought to be caused by the consumption of food or water. Food borne infection involves the ingestion of the pathogen followed by growth in the host, including tissue invasion the release of toxins. The area of diary microbiology is large and diverse. The bacteria in diary products may cause disease or spoilage. Some bacteria may be specifically added to milk for fermentation to produce products like yoghurt and cheese.

Evaluation

Multiple choice questions

1. The primary sources of micro organisms in food include _____

   i. Soil and water

   ii. Food utensils

   iii. Food handlers

   iv. Air and dust

   a. (i) and (ii)

   b. (ii), (iii)

   c. (i), (ii), (iii) and (iv)

   d. None of the above

2. The micro organisms grow best at PH value around _____

   a. 4.0 b. 7.0 c. 3.4 d. 9.2

3. The aw of most fresh food is above _____

   a. 0.99 b. 0.88 c. 0.77 d. 0.66

4. The minimum value of aw for the growth of the micro organisms in foods should be around _____

   a. 0.99 b. 0.86 c. 0.78 d. 0.50

5. Choose mismatched pair:

   Asepsis – Keeping out of Micro organisms

   Filteration – Removal of Micro organisms

   Heat (or) Radiation – Killing the Micro organisms

   Prevention of Damage – Blanching

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